

MAT8034 Final Project Guidelines

2026 Spring

Overview

One of MAT8034's main goals is to prepare you to apply machine learning algorithms to real-world tasks, or to leave you well-qualified to start machine learning or AI research. The final project is intended to start you in these directions.

- **Team Size:** 1 to 3 members are allowed
- **Programming Language:** No restriction
- **Project Topic:** Any ML related topics is acceptable

Project Category

There are three different types offered here, and you are free to choose one depending on your team's background and interests. Feel free to brainstorm with your TAs for inspiration.

1. Survey + Reproduction

- **Requirement**
 - **Comprehensive Review:** Select a specific ML subfield, map out its development trajectory, summarize the development context, comparing different methodologies. You must analyze the pros and cons of these methods and discuss future trends. Your survey must include at least **5 papers** published **within the last 3 years**.
 - **Replication:** Select 1 core paper from your survey and rigorously **reproduce** its main experimental results.
- **Helpful Resources & Tools**
 - Knowledge Bases: PapersWithCode , arXiv
 - Trend Tracking: Check the Hugging Face Daily Papers/Github trending to see what the open-source community is actively discussing.
 - Top-Tier Conferences: Search for SOTA papers in recent proceedings of NeurIPS, ICML, ICLR (General/Deep Learning); CVPR, ICCV (Vision); ACL, EMNLP (NLP); and KDD (Data Mining).

2. Applied / Engineering

- **Requirement**

Solve a practical problem by putting algorithms into production.

Select a real-world dataset (e.g., public datasets, competition data). You must **clearly formulate the problem**, perform thorough data preprocessing/cleaning, and conduct a rigorous comparison of **at least three different methods**.

- **Helpful Resources & Tools**

- Kaggle: The largest platform for structured/unstructured data competitions.
- KDD Cup: Annual premier data mining competition.
- Conference workshop
- DrivenData: Focuses on data science projects for social good and real-world impact.
- AICrowd & CodaLab: Excellent for specialized RL, vision, and NLP challenges.

3. Research

- **Requirements**

Push the boundaries of current knowledge by proposing a novel idea.

Clearly define your algorithmic or theoretical innovation. You must provide rigorous empirical or theoretical validation, directly comparing your work against **State-of-the-Art (SOTA) baselines from the past 3 years**.

Topics of interest include (but are not limited to)

- **General Machine Learning:** Active learning, clustering, online learning, ranking, supervised, semi- and self-supervised learning, time series analysis, etc.
- **Deep Learning:** Architectures, generative models, theory, etc.
- **Evaluation:** Methodology, meta-studies, replicability and validity, human-in-the-loop, etc.
- **Theory of Machine Learning:** Statistical learning theory, bandits, game theory, decision theory, etc.
- **Machine Learning Systems:** Improved implementation and scalability, hardware, libraries, distributed methods, etc.
- **Optimization:** Convex and non-convex optimization, matrix/tensor methods, stochastic, online, non-smooth, composite, etc.
- **Probabilistic Methods:** Bayesian methods, graphical models, Monte Carlo methods, etc.
- **Reinforcement Learning:** Decision and control, planning, hierarchical RL, robotics, etc.
- **Trustworthy Machine Learning:** Reliability, causality, fairness, interpretability, privacy, robustness, safety, etc.
- **Application-Driven Machine Learning:** Innovative techniques, problems, and datasets driven by the needs of end-users in healthcare, physical sciences,

biosciences, social sciences, sustainability, climate, etc.

Project Components & Submission

Our project will consist of three parts: Proposal, Presentation, and Final Report. We'll use Blackboard for submission of presentation slides and final report.

Part I: Proposal

- **Format:** Your proposal should give the title of the project, the category, the full names of all of your team members with SID and a 80-150 words abstract of what you plan to do.
- **Content:** Include your Motivation, Method, and Intended experiments. Providing links to one relevant dataset and prior research is optional but valuable.

Part II: Presentation

- **Format:** You are required to submit an exhibition PPT or PDF file (e.g. generated by Beamer) based on your project.

Part III: Final Report

- **Format:** The final project report should not exceed **8 pages** and should be formatted according to the **template**. We will allow for extra pages containing only references.
- **Content:** Describe the problem setting, methods, theoretical and empirical results. Please include a section that briefly describes what each team member worked on and contributed to the project. (Requires contributions to be labeled by percentage, otherwise it defaults to equal).
- **Code:** Please include a zip file with the codes for your final project. The zip file should not exceed 5MB.

Evaluation Criteria

The project score accounts for **40** points of the final course grade. The project is evaluated across three core dimensions:

1. **Technical Quality:** Are the methods reasonable? Is the analysis in-depth?...
2. **Originality & Effort:** Did the authors add their own data processing, methods, or robust analysis? ...
3. **Presentation:** Clarity and rigor in explaining context, methods, and results.

Important Dates (Spring 2026)

Stage	Deadline (before 23:59)	Deliverable
Team Formation	March 30, 2026	Team information (via shared document)
Proposal	April 10, 2026	Abstract (via shared document)
Presentation	May 25-June 1, 2026	PowerPoint/PDF (via Blackboard)
Final Report	June 7, 2026	PDF + ZIP code (via Blackboard)

FAQs

- Should the project use only methods taught in the class?

No, we don't restrict you to only use methods/topics/problems taught in class. That said, you can always consult a TA if you are unsure about any method or problem statement.

- Is it okay to use a dataset that is not public?

We don't mind you using a dataset that is not public, as long as you have the required permissions to use it. We don't require you to share the dataset either as long as you can accurately describe it in the final report.

- What if two teams end up working on the same project?

If two or more teams have selected the same or similar projects, they will not be recognized as plagiarists as long as the teams have different ways of thinking about how to carry out their work. Alternatively, these teams could collaborate, but would need to ensure that each team worked on different sub-problems.